

Notes: Froot, Scharfstein & Stein (JF, 1993)

Risk Management: Coordinating Corporate Investment and Financing Policies

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1 Big picture

- **Question.** Why should a corporation hedge, and how should a corporation choose the amount and form of hedging when financing frictions interact with investment opportunities?
- **Core friction.** External finance is more costly than internal funds. If cash flow falls in states where the firm has valuable investment opportunities, the firm may cut investment or raise expensive outside finance.
- **Main mechanism.** Hedging creates value by aligning the supply of internal funds with the demand for investment funds. The object to stabilize is not necessarily firm value or cash flow; it is the *shadow value of internal funds*.
- **Main result.** Hedging is valuable when profits are concave in internal wealth. This requires both decreasing marginal returns to investment and a positive effect of internal wealth on investment.
- **Strategic result.** Optimal hedging generally does not mean complete insulation of firm value from hedgeable risks. The firm may hedge less than one-for-one, more than one-for-one, or even take additional exposure, depending on how the risk affects future investment and financing opportunities.
- **Instrument choice.** Linear instruments such as futures and forwards are not always enough. Options are valuable when the desired hedge ratio varies across states of the world.
- **Multinational implication.** Currency hedging should depend on whether exchange rates affect foreign revenues, foreign investment costs, and the desired allocation of investment across countries.
- **Empirical implication.** If the firm is optimally hedged, the marginal value of internal funds should not covary with the risk variable being hedged.
- **Takeaway.** Risk management is a corporate-finance policy, not merely a trading or risk-reduction policy. Its purpose is to coordinate investment and financing decisions under costly external finance.

1.1 Incremental contribution of the paper

- **From “why hedge” to “how hedge.”** Earlier theories often justified hedging through taxes, managerial risk aversion, or financial distress. This paper’s incremental contribution is to turn the capital-market-imperfection rationale into a theory of *hedging strategy*: which risks to hedge, how much to hedge, and which instruments to use.
- **Coordination of investment and financing.** The paper explicitly links risk management to the firm’s investment schedule and external-finance supply curve. Hedging is not valuable because risk is bad per se; it is valuable because cash-flow risk distorts investment and financing choices.
- **Shadow-value principle.** The key normative object is the marginal value of internal wealth, P_w . Optimal hedging makes this shadow value insensitive to hedgeable shocks. This is sharper than the usual prescription to reduce earnings volatility or stock-price volatility.
- **State-contingent hedge ratios.** The paper shows that the desired hedge depends on the state-contingent behavior of investment opportunities and financing capacity. This naturally leads to incomplete hedging, overhedging, or negative hedging.
- **Instrument design.** The paper explains why options, forwards, futures, swaps, and indexed debt differ economically. Options are useful when the desired cash-transfer schedule is nonlinear; futures can create margin-call liquidity risk; forwards avoid margin calls but create credit risk.
- **Empirical discipline.** The paper proposes a testable restriction: if hedging is optimal, firm value should not be more sensitive to cash flow in states where hedgeable risk makes internal funds especially valuable. This anticipates later empirical work on cash-flow sensitivity, financing constraints, and corporate hedging.

2 Model primitives and measurement

2.1 Firm investment and financing

The firm has internal resources w and chooses investment I . External financing is

$$e = I - w.$$

Investment produces expected output $f(I)$, where $f_I > 0$ and $f_{II} < 0$. Net present value before financing costs is

$$F(I) = f(I) - I.$$

External finance imposes deadweight costs

$$C = C(e), \quad C_e > 0.$$

Interpretation: The model captures the central empirical idea behind financing constraints: a dollar of internal wealth is not equivalent to a dollar of outside finance when outside finance carries deadweight costs.

2.2 Why hedging enters the model

The firm faces uncertainty in internal wealth w . The paper assumes, as a benchmark, that the risk in w is marketable and hedgeable. Hedging does not change expected wealth, but it changes the distribution of wealth across states.

Interpretation: Hedging matters only because the firm's value is nonlinear in internal wealth. If $P(w)$ were linear, moving cash across states would not affect expected firm value.

2.3 Investment opportunities

In the richer model, investment opportunities can change with the same risk variable that affects internal funds. The paper represents this through a productivity term

$$F(I) = \theta f(I) - I,$$

where θ increases or decreases with the hedgeable risk variable.

Interpretation: If cash flows are high exactly when investment opportunities are good, the firm already has a natural hedge. It should hedge less. If cash flows are low when investment opportunities are good, the firm should hedge more.

2.4 Financing opportunities

The cost of external finance can also vary with the risk variable. The paper writes this as

$$C = C(e, \phi),$$

where ϕ summarizes collateral values or the pledgeable value of assets in place.

Interpretation: If collateral values fall in bad cash-flow states, the firm needs extra hedging. Hedging should protect not only operating cash flows but also the ability to raise external finance.

2.5 Hedging instruments

The paper distinguishes between:

- **linear hedges:** futures and forwards, which create a constant exposure across states;
- **nonlinear hedges:** options and option-like contracts, which create state-contingent exposure;
- **forward-like contracts:** swaps and indexed debt, which avoid interim margin calls but may create counterparty credit risk.

Interpretation: The best instrument is determined by the desired state-contingent cash-flow profile. The paper is therefore about hedge design, not merely hedge quantity.

3 Models and theoretical specifications

3.1 Basic investment-financing problem

The firm chooses investment to maximize

$$P(w) = \max_I \{F(I) - C(I - w)\}.$$

The first-order condition is

$$f_I(I) - 1 = C_e(I - w).$$

Interpretation: The first-best condition would be $f_I = 1$. With costly external finance, the firm chooses $f_I > 1$, meaning that it underinvests relative to first-best. The wedge is the marginal cost of outside finance.

3.2 Why hedging raises value

The paper shows that expected firm value rises from reducing the volatility of w when $P(w)$ is concave. Using the first-order condition, the curvature can be written as

$$P_{ww} = f_{II} \frac{dI^*}{dw}.$$

Thus hedging is valuable if

$$f_{II} < 0 \quad \text{and} \quad \frac{dI^*}{dw} > 0.$$

Interpretation: Two things must be true. First, investment must have decreasing marginal returns. Second, internal funds must relax an investment constraint. If either condition fails, cash-flow smoothing has little or no value.

3.3 Reduced-form interaction of investment and financing

A more explicit expression is

$$P_{ww} = \frac{f_{II} C_{ee}}{f_{II} - C_{ee}}.$$

Interpretation: The concavity of firm value comes from the interaction between the concavity of the production technology and the convexity of the external-finance cost function. Hedging is therefore most valuable when projects are concave and financing costs rise sharply with external finance.

3.4 Costly-state-verification foundation

The paper grounds the reduced-form financing cost in a costly-state-verification model. The firm invests I and raises outside financing. Assets in place produce random cash flow x with density $g(x)$ and distribution $G(x)$. Outside investors can verify cash flow at monitoring cost c . The optimal contract is debt with promised repayment D .

The firm solves

$$\max_{I,D} \left\{ f(I) + \int_D^\infty (x - D)g(x) dx \right\}$$

subject to investor break-even:

$$\int_0^D (x - c)g(x) dx + \int_D^\infty Dg(x) dx \geq I - w.$$

The first-order conditions imply

$$f_I = \frac{1 - G(D)}{1 - G(D) - cG(D)} \geq 1.$$

Expected deadweight costs are

$$C(e) = cG(D).$$

Interpretation: This section shows that the convex external-finance cost is not ad hoc. It can arise from standard debt contracts when monitoring or bankruptcy is costly. Additional financing raises the repayment promise, which increases the probability of costly verification.

3.5 Changing investment opportunities and the linear hedge ratio

Let ε be the hedgeable risk variable, and let the firm choose hedge ratio h so that

$$w = w_0[h + (1 - h)\varepsilon].$$

A linear hedge ratio of $h = 1$ fully insulates internal wealth from ε . The first-order condition for the optimal hedge is

$$E \left[P_w \frac{dw}{dh} \right] = 0,$$

which becomes

$$\text{cov}(P_w, \varepsilon) = 0.$$

Interpretation: The optimal hedge does not necessarily make cash flow constant. It makes the marginal value of cash constant with respect to the risk variable. If cash is most valuable in low- ε states, the firm should transfer cash toward those states. If cash is more valuable in high- ε states, the firm should retain or increase exposure.

3.6 Investment-opportunity correlation

When investment opportunities are positively correlated with cash flows, the optimal hedge ratio is below one. When investment opportunities are negatively correlated with cash flows, the optimal hedge ratio can exceed one.

Interpretation: Oil companies are the key example. Low oil prices reduce cash flows but also reduce the attractiveness of oil exploration. That natural matching lowers the value of hedging oil-price risk. A gold firm whose investment opportunities are less sensitive to gold prices should hedge more.

3.7 Changing financing opportunities

If collateral values or financing capacity move with the hedgeable variable, the firm may need to hedge more. In the paper's notation, if marginal financing costs fall when the risk variable is high, the optimal hedge can satisfy

$$h^* > 1.$$

Interpretation: Overhedging is not irrational speculation. It is a way of transferring internal funds into states where external finance becomes especially expensive.

3.8 Multinational currency hedging

The multinational can invest at home and abroad. Profits are

$$P(w) = f_H(I_H) + \theta f_A(I_A) - I_H - \gamma I_A - C(e),$$

where θ captures exchange-rate exposure of foreign revenues and γ captures exchange-rate exposure of foreign investment costs. External financing needs are

$$e = I_H + \gamma I_A - w.$$

A key special case is $\alpha = \beta$, where foreign revenues and foreign investment costs have the same exchange-rate exposure. Then the firm can hedge so as to lock in planned investment quantities across countries.

Interpretation: Currency hedging depends on real-side exposures. A firm should not hedge exchange rates solely because its accounting cash flow is exposed; it should hedge based on how exchange rates change the desired allocation of investment across countries.

3.9 Nonlinear hedging and options

With complete markets, the firm chooses state-contingent internal wealth $w(\varepsilon)$:

$$\max_{w(\varepsilon)} \int P(\varepsilon, w(\varepsilon)) p(\varepsilon) d\varepsilon$$

subject to fair pricing:

$$\int w(\varepsilon) p(\varepsilon) d\varepsilon = w_0.$$

The first-order condition is

$$P_w = \lambda.$$

Interpretation: The optimal nonlinear hedge equalizes the shadow value of internal wealth across states. Options help because they can adjust exposure differently in extreme states than in moderate states.

3.10 Options density and state-contingent hedge ratios

The first derivative of the desired wealth profile, $dw^*/d\varepsilon$, gives desired exposure to the risk variable. The second derivative, $d^2w^*/d\varepsilon^2$, determines the density of options at different strike prices.

Interpretation: Options are not valuable because they are exotic. They are valuable because the optimal cash-transfer schedule is often nonlinear. If the firm needs protection mainly in extreme bad states, out-of-the-money puts can dominate a simple futures hedge.

4 Identification and theoretical strategy

4.1 Nature of the paper

This is a theoretical and prescriptive paper, not an empirical causal design. The “identification” of the mechanism comes from the model’s equilibrium logic: costly external finance makes the marginal value of internal funds state dependent; hedging creates value by reducing that state dependence.

4.2 Step 1: establish a reason for hedging

The paper first shows that if internal funds affect investment and investment has diminishing returns, firm value is concave in internal wealth. Therefore, cash-flow smoothing increases expected value.

Interpretation: This is the central departure from Modigliani-Miller irrelevance. Investors could hedge their personal portfolios, but they cannot costlessly restore the firm’s missing internal funds when external finance is costly.

4.3 Step 2: move from full hedging to optimal hedging

Once investment opportunities or financing opportunities are stochastic, full cash-flow hedging is no longer generally optimal. The first-order condition becomes

$$\text{cov}(P_w, \varepsilon) = 0.$$

Interpretation: This condition disciplines the theory. If a risk naturally provides cash when cash is most valuable, hedging it away can destroy value. If a risk removes cash when cash is most valuable, the firm should hedge more aggressively.

4.4 Step 3: derive instrument implications

Linear instruments can implement only constant exposures. If the desired exposure varies by state, nonlinear instruments are required. This leads to a theory of why options should be used by firms whose desired hedge ratios are state dependent.

Interpretation: The instrument choice is endogenous to investment and financing opportunities. Risk management policy cannot be separated from capital budgeting and capital structure.

4.5 Step 4: connect to empirical tests

The paper proposes an empirical specification:

$$P_{t,i} = a + w_{t,i}(a_1 + a_2 E_t) + a_3 E_t + v_{t,i},$$

where P is firm value, w is contemporaneous cash flow, and E is the risk variable. Optimal hedging implies

$$a_2 = 0.$$

Interpretation: The cross term a_2 measures whether the marginal value of internal funds depends on the hedgeable risk variable. If $a_2 \neq 0$, the firm could improve value by changing its hedge.

4.6 Threats and assumptions

- The risk variable must be plausibly exogenous to other omitted determinants of firm value.
- The coefficient a_2 can be biased if cash flow predicts future profitability differently across risk states.
- Firm value may respond to cash flow for reasons unrelated to liquidity constraints.
- Investment regressions with Tobin's Q are an alternative, but they measure quantity rather than value of investment.
- Direct hedging data are difficult to obtain because derivatives are often off-balance-sheet.

Interpretation: The proposed test is conceptually sharp but empirically demanding. Later empirical work on investment-cash flow sensitivity and financing constraints can be interpreted through this lens.

5 Table

5.1 Table I: Hypothetical Hedging Strategies and Investment Spending

Read:

- The example has three equally likely states with first-best investment of 6, 9, and 15.
- Without hedging, internal funds are 5, 10, and 15. The firm underinvests in state 1 and has excess cash in state 2.
- A futures hedge can transfer cash from state 3 to state 1, producing 6, 10, and 14. This protects the low state but still fails to match first-best investment in state 3.
- Options complete the state-contingent cash-transfer schedule: one unit of a state-1 payoff and two units of a state-3 payoff, costing one unit in total, produce 6, 9, and 15.

Interpretation: The table demonstrates why linear hedges can be insufficient. A futures hedge moves cash along a straight line across states; options permit more precise state-by-state transfers of cash.

Economic message: Options are valuable when the value-maximizing hedge ratio is not constant across states. They can direct cash to exactly the states where investment would otherwise be distorted.

Table I
Hypothetical Hedging Strategies and Investment Spending
(with Initial Wealth of 10)

State	Probability	Optimal Investment Spending	Net Funds Available for Investment			
			No Hedging (1)	Optimal Futures Hedge (2)	Payoffs to First-Best Options (3)	First-Best Hedge with Options (2) + (3) - cost
1	1/3	6	5	6	1	$6 + 1 - 1 = 6$
2	1/3	9	10	10	0	$10 + 0 - 1 = 9$
3	1/3	15	15	14	2	$14 + 2 - 1 = 15$
Total cost of options				$-1/3 - 2/3 = -1$		

Table I: Hypothetical Hedging Strategies and Investment Spending

6 Interpretive synthesis

6.1 Why this paper changed the corporate hedging literature

The paper reframes risk management away from the variance of firm value and toward the interaction between liquidity and investment. The key insight is that shareholders care about whether the firm can fund positive-NPV projects. Hedging is valuable when it improves that funding ability.

6.2 Why complete hedging is usually wrong

Complete hedging of cash flow is optimal only in special cases: nonstochastic investment opportunities and fixed financing opportunities. Once the risk variable also changes project profitability or external financing capacity, the firm should preserve some exposure or create more exposure.

6.3 Why hedging is part of capital budgeting

Capital budgeting asks which projects should be undertaken. Hedging changes the state-contingent availability of funds to undertake those projects. Therefore, the hurdle rate and the hedge policy are not independent decisions when external finance is costly.

6.4 Why derivative choice matters

The choice between futures, forwards, options, swaps, and indexed debt is not merely a question of contract mechanics. Each contract creates a different state-contingent cash-flow profile. The firm should choose the contract that best equalizes the shadow value of internal funds across states.

7 Short summary

- The paper develops a theory of corporate risk management based on costly external finance.
- Hedging adds value by helping firms fund valuable investment opportunities with internal funds.
- The basic model shows that hedging is valuable when firm value is concave in internal wealth.
- This concavity arises when investment has decreasing marginal returns and internal funds affect investment.
- A costly-state-verification model provides a microfoundation for convex external-finance costs.
- When investment opportunities are correlated with cash flows, full hedging is generally not optimal.
- When financing opportunities are correlated with cash flows, overhedging can be optimal.
- Multinational currency hedging depends on exchange-rate exposure of revenues and investment costs.
- Options are useful when the desired hedge ratio varies across states.
- The empirical restriction is that optimal hedging makes the marginal value of internal funds uncorrelated with the hedgeable risk variable.

8 Conclusion

8.1 Core conclusion

Froot, Scharfstein, and Stein show that corporate risk management should be designed to coordinate investment and financing policies. Hedging is valuable when it reduces costly variation in the availability of internal funds relative to investment needs.

8.2 Economic intuition

A firm with costly external finance may cut investment when cash flow is low, even if the project has positive NPV. A hedge can transfer cash into those states, preventing inefficient underinvestment. But if low cash-flow states also have poor investment opportunities, full hedging is unnecessary. The optimal hedge follows the shadow value of funds, not the volatility of cash flow.

8.3 Practical implications

- Firms with valuable growth opportunities and high external-financing costs should hedge more.
- Firms whose investment opportunities naturally covary with cash flows should hedge less.
- Firms whose collateral value falls in bad cash-flow states may need to overhedge.
- Multinationals should base currency hedging on real investment exposure, not accounting exposure alone.
- Options are useful when the firm needs cash protection in extreme states more than in ordinary states.

8.4 What the paper does not claim

The paper does not claim that every firm should hedge or that hedging should eliminate all risk. It does not provide a fully estimated empirical model of corporate hedging. Its contribution is a framework for deciding what hedging policy should do when investment and financing frictions matter.

8.5 Final takeaway

The central lesson is simple but powerful: the purpose of risk management is not to smooth reported earnings or stabilize stock prices mechanically. The purpose is to make sure that the firm has cash when cash is most valuable.